BERT: Bidirectional Encoder Representations from Transformers

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In this video you will be introduced to BERT, a method and toolkit from Google Labs for understanding meaning relationships between sentences, and used in tasks which involve measuring meaning similarity between sentences.

Devlin et al 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of NAACL'2019 North American Chapter of the Association for Computational Linguistics (Best Paper award)

https://www.aclweb.org/anthology/N19-1423.pdf

by Jacob Devlin, Ming-Wei Chang, Kenton Lee, Kristina Toutanova, Google AI Language Research
- can University researchers compete with Google, MS, FB etc?

Overview: BERT

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Pre-trained representations from unlabeled text, then fine-tuned

Unsupervised learning of morphology, syntax and semantics

WordPiece unsupervised ML of text segmentation

Predicting sentence-meaning from wordPiece meanings

Assume adjacent sentences have related meanings

Results: BERT fine-tuned on human-labelled NLP data-sets

“Both BERTBASE and BERTLARGE outperform all systems on all tasks by a substantial margin … the same pre-trained model can successfully tackle a broad set of NLP tasks”

Also: Ablation Studies, References, Appendices, Questions …

BERT: pre-trained representations from unlabeled text, then fine-tuned

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BERT stands for Bidirectional Encoder Representations from Transformers – language representation models, DATA not software, though you also need software to use BERT models

“pre-trained representations from unlabeled text” – a model of general English words and grammar “For pre-training we use the BooksCorpus (800M words) and English Wikipedia (2,500M words)”

“can be fine-tuned to create models for a wide range of tasks” – tweak for specific language task “state-of-the-art results on eleven natural language processing tasks” - GLUE, SQuAD etc.

“The code and pre-trained models are available at https://github.com/google-research/bert ”

Unsupervised learning of morphology, syntax and semantics

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BERT Transformer NN maps pair of texts onto a class.
Train: show BERT many examples: sentence+sentence = YorN
Test: show BERT new sentence+sentence; BERT predicts YorN

BERT learns (sentence+sentence+class) without linguistic theory of morphology (word structure), syntax, semantics (?)

Instead, unsupervised (or semi-supervised) learning of morphology, syntax and semantics from a large text corpus

(like MorphoChallenge to segment words into morphemes, word2vec to learn semantics-vectors for words/morphemes, plus learning if sentence-pairs have similar meanings)

WordPiece unsupervised machine learning of text segmentation

First, segment each sentence: not words but WordPieces

Linguistic theory: words and morphemes

BUT how to handle rare words and OutOfVocabulary words?

“We use WordPiece embeddings with a 30,000 token vocabulary”

WordPiece is an unsupervised ML morphological analyser: common words are kept as Pieces, rare/unknown words are chopped into common Pieces (like MorphoChallenge)

So: no rare, OOV Words; only common WordPieces

Then each WordPiece is mapped to an Embedding, a vector representing concordance contexts (like word2vec)

Predicting sentence-meaning from wordPiece meanings

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Assumes 25 wordPieces per sentence; if not, truncate or fill.

Semantics of a sentence: join meaning vectors of wordPieces.

“We do not use traditional left-to-right or right-to-left language models to pre-train BERT ... we simply mask some percentage of the input tokens at random, and then predict those masked tokens. This is a “masked LM” (MLM), although it is often referred to as a Cloze task”

Assume adjacent sentences have related meanings

Novel idea: in a text, each sentence is related in meaning to the next sentence: sentence-meaning ”bigram model”

“human-labelled”? Assumes author write coherent text, where each sentence “follows on from” the previous sentence

“We pre-train for a next sentence prediction (NSP) task: in training sentences A and B, 50% of the time B is the actual next sentence that follows A (labeled as IsNext), and 50% of the time it is a random sentence from the corpus (labeled as NotNext). ... the input sentence A and sentence B from pre-training are analogous to (1) sentence pairs in paraphrasing, (2) hypothesis-premise pairs in entailment, (3) question-passage pairs in question answering, and (4) text-0 pairs in text classification ... fed into an output layer for classification”

Experiments: BERT fine-tuned results on human-labelled NLP data-sets

GLUE General Language Understanding Evaluation tasks:

SST-2: Given a movie review, predict sentiment: + or –

CoLA: Given a sentence, is it linguistically “acceptable”? YorN

STS-B, MRPC: Given a pair of sentences, do they have similar meaning? YorN

QQP: Given a pair of questions on Quora, are the two questions semantically equivalent? YorN

QNLI: Given a question and a sentence, does the sentence contain the correct answer? YorN

MNLI, RTE: Given first sentence, is the second sentence an entailment, contradiction, or neutral?

Experiments on large human-labelled data-sets

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SQuAD Stanford Question Answering Dataset - 100,000 question/answer-passage pairs Given a question, predict the answer text span in a Wikipedia passage.

SWAG Situations With Adversarial Generations - 100,000 sentence-completion examples Given a sentence, choose the most plausible continuation from 4 choices.

“Both BERTBASE and BERTLARGE outperform all systems on all tasks by a substantial margin; BERTLARGE significantly outperforms BERTBASE across all tasks.” (NN layers: BASE 12, LARGE 24)

In addition…

Ablation Studies – experiments to cut out components to see how this affects performance

Conclusion – “deep *bidirectional* architectures allow the same pre-trained model to successfully tackle a broad set of NLP tasks”

References - a survey of state-of-the-art NLP research!

Appendices- “implementation details”; more details on experiments and ablation studies

Questions about BERT

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The authors pack a LOT into 9 pages (+refs+appendices)
BUT some issues are not specified:

Implementation details - code examples to copy and try (URL)
How are sentences with different lengths encoded as inputs?
BIG computing resources required – what works on a laptop?
Examples of good and **bad** results (and analysis/explanation)

What sort of data/tasks do NOT suit BERT?
English only – how does this transfer to other languages?
Assumes WordPiece tokenization; what about Multi Word Expressions, and languages with rich morphology?

Can we “understand” how BERT works? It seems counter-intuitive: **BERT** learns (sentence+sentence+class) without linguistic theory: rules of morphology, syntax, semantics

Summary: BERT

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